OSHE MODULE 4 – MAIN POINTS SUMMARY

Health Considerations at Workplace & PPE

Q1 – Health Problems at Workplace

- **Infectious Diseases (Airborne):** TB, Influenza, Measles, Chickenpox, SARS, COVID-19
- **Infectious Diseases (Contact):** Hepatitis A, Dysentery, Typhoid, E. coli, Salmonellosis
- **Occupational Respiratory:** Asthma, COPD, Silicosis, Coal Workers' Pneumoconiosis (CWP)
- **Occupational Skin Diseases (OSD):** Irritant/Allergic Dermatitis, Skin Infections, Skin Cancer
- **Musculoskeletal Disorders (MSDs):** RSI, Back Pain, Neck Pain, Carpal Tunnel Syndrome, Hernia
- **Psychosocial/Mental:** Stress, Burnout, Depression, Anxiety → low morale, hypertension
- **Occupational Cancer:** Long-term exposure to carcinogenic chemicals/fumes
- **Lifestyle Diseases:** Obesity, Type 2 Diabetes, Hypertension, CVD (sedentary work)
- **Prevention:** Hygiene, ventilation, regular check-ups, vaccination, PPE, stress programs

Q2 – Classification of RSI & Associated Disorders

- **Type I RSI (Specific):** Recognizable condition; swelling/inflammation present; medically diagnosable
- **Type II RSI (Non-Specific):** No visible inflammation; pain, stiffness, fatigue; hard to diagnose

Associated Disorders

- **Tendinitis:** Tendon inflammation; shoulders, elbows, wrists
- **Tenosynovitis:** Tendon sheath inflammation; common in typists & assembly workers
- **Carpal Tunnel Syndrome:** Median nerve compression; numbness, tingling, weak grip
- **Bursitis:** Inflamed bursae; shoulders, elbows, knees; swelling & pain
- **Epicondylitis (Tennis Elbow):** Tendon inflammation around elbow; repeated gripping/twisting
- **Neck & Shoulder Strain:** Poor posture, prolonged computer use; stiffness & pain
- **Back Disorders:** Improper lifting/sitting; lower back pain, reduced mobility
- **Prevention:** Ergonomic posture, rest breaks, ergonomic tools, task rotation, training

Q3 – Measures to Control Occupational Health Risks

- **Engineering Controls:** Ventilation systems, humidity <60%, splash guards, enclose/isolate hazardous processes
- **Administrative Controls:** Employee rotation, limit exposure, health monitoring, sick-leave policy, OHS training
- **Workplace Hygiene:** Handwash ≥15 sec, clean/disinfect surfaces, proper waste disposal, clean water
- **Vaccination & Surveillance:** Vaccines, regular check-ups, occupational disease screening, medical records
- **PPE:** Helmets, goggles, gloves, ear plugs, respirators, safety shoes, aprons, harnesses
- **Ergonomic Measures:** Ergonomic workstations, correct posture, avoid repetition, rest breaks, adjustable furniture
- **Mental Health:** Reduce overload, clear roles, counseling, work-life balance
- **Emergency Preparedness:** Response plans, first aid/CPR training, drills, first aid kits

Q4 – Byssinosis & Pneumoconiosis

i) Byssinosis (Brown Lung Disease)

- **Cause:** Inhalation of cotton, flax, hemp, or jute dust – textile industry workers
- **Symptoms:** Chest tightness (Monday mornings), breathlessness, persistent cough, wheezing, fatigue, reduced lung function
- **Precautions:** Ventilation & dust extraction, worker rotation, health exams, dust masks/respirators, regular cleaning

ii) Pneumoconiosis

- **Cause:** Inhalation of coal dust, silica, asbestos → lung fibrosis (scarring); includes Silicosis & CWP
- **Symptoms:** Persistent cough, breathlessness, chest pain, wheezing, excess sputum, reduced lung capacity, fatigue
- **Precautions:** Local exhaust ventilation, wet drilling, dust suppression, medical check-ups, N95 masks, lung function tests, chest X-rays

Q5 – Common Workplace Diseases & Health Emergency Planning

Common Diseases

- **Airborne Infectious:** TB, Influenza, SARS, Measles, Chickenpox, Meningitis, Smallpox
- **Contact Infectious:** Hepatitis A, Dysentery, Typhoid, Salmonellosis, E. coli, MRSA
- **Biological Risk:** COVID-19, HIV/AIDS, Hepatitis B & C, Avian Influenza
- **Occupational:** Skin diseases, Respiratory diseases, MSDs, Psychosocial disorders, Occupational Cancer

Health Emergency

- **Definition:** Sudden medical situation threatening health/life requiring immediate response
- **Examples:** Cardiac arrest, choking, severe bleeding, burns, poisoning, gas leakage

Planning for Health Emergency

- **Training:** First aid/CPR training, emergency drills, clearly assigned roles
- **Life-Threatening Response:** Life support within 3–4 minutes (cardiac arrest, choking, severe bleeding)
- **Non-Life-Threatening:** First aid within 15 minutes (minor burns, cuts)
- **Check–Call–Care:** Check scene & victim → Call emergency services → Care (first aid/CPR until help arrives)
- **Minor Injuries:** Sterile dressing, pressure to stop bleeding, cool burns with water
- **Serious:** Heimlich maneuver for choking; CPR = 30 compressions + 2 rescue breaths

Q6 – Health Problems: Hospitals, Nursing Homes & Entertainment Places

Hospitals & Nursing Homes

- **Infectious Diseases:** TB, COVID-19, Influenza, Hepatitis B/C, HIV/AIDS, Meningitis
- **Biological Hazards:** Needle-stick injuries, contact with contaminated waste/fluids
- **MSDs:** Back/neck pain, joint disorders, RSI – from lifting patients, prolonged standing
- **Mental Health:** Stress, burnout, fatigue, anxiety – heavy workload, emergencies, long hours

Entertainment Places

- **Communicable Diseases:** Influenza, COVID-19, Measles, Chickenpox – overcrowding, poor ventilation
- **Noise-Related:** Hearing loss, tinnitus, headaches – loud music & sound systems
- **Respiratory:** Allergies, asthma, breathing difficulties – dust, poor air quality
- **Accident/Injury Risks:** Slips, fire accidents, crowd injuries, electrical hazards

Management & Control

- **Engineering:** Ventilation, air filtration, humidity <60%, sanitation, barriers
- **Administrative:** Health monitoring, infection-control policies, limit overcrowding, rotate staff
- **Hygiene:** Disinfection, waste disposal, handwashing ≥15 sec, clean toilets
- **Vaccination + PPE + Emergency:** Immunization; masks, gloves, goggles; first aid/CPR drills

Q7 – Types of Personal Protective Equipment (PPE)

Non-Respiratory PPE

- **Head (Helmets):** Class G (general), Class E (electrical), Class C (conductive)
- **Eye & Face:** Safety glasses, goggles, face shields
- **Hands & Arms:** Nitrile/latex gloves, Kevlar cut-resistant, rubber insulating gloves
- **Hearing:** Ear plugs, ear muffs – prevents Noise-Induced Hearing Loss (NIHL)
- **Foot & Leg:** Steel-toed boots, metatarsal guards, slip-resistant shoes
- **Body:** Leather/PVC/asbestos aprons, high-visibility vests, full-body suits
- **Fall Protection:** Safety belts, harnesses, lifelines, lanyards – required at heights >2 m; inspect every 6 months

Respiratory PPE

- **N95 Respirator:** Filters ≥95% airborne particles; dust & pathogens
- **Air-Purifying Respirators:** Half-face/full-face; replaceable filters for chemical vapors
- **SCBA (Supplied-Air):** External clean air; used in O_2-deficient or emergency environments

Characteristics of Good PPE

- Adequate protection, comfortable & lightweight, no movement restriction, durable & easy to maintain, meets safety standards, attractive to users

Q8 – Importance & Need of PPE at Workplace

Need

- **Protection from hazards:** Physical, chemical, biological, electrical, mechanical
- **When other controls insufficient:** Last line of defense when engineering/admin controls fall short
- **Prevent occupational diseases:** Respiratory diseases, skin disorders, hearing loss, eye injuries
- **Emergency safety:** Fires, chemical spills, gas leaks, rescue operations
- **Legal compliance:** Meets Factories Act and OHS regulatory requirements

Importance (Body Protection)

- **Head:** Helmets – falling objects, impact, electrical hazards
- **Eyes & Face:** Goggles/shields – flying particles, dust, chemical splashes, radiation
- **Respiratory:** N95/respirators – dust, vapors, pathogens → prevents silicosis, asthma
- **Hands:** Gloves – cuts, chemical burns, electrical shocks, dermatitis
- **Hearing:** Ear plugs/muffs – excessive noise → prevents NIHL & tinnitus
- **Feet:** Safety boots – crushing, punctures, slips
- **Body:** Aprons/suits – chemicals, heat, contamination, visibility

- **Fall:** Belts/harnesses/lifelines – fatal fall prevention at heights >2 m

Q9 – Factors Affecting Selection of PPE

- **Nature of Hazard:** Specific PPE matched to specific hazard (helmet→falling objects, goggles→chemicals)
- **Adequate Protection:** Must effectively reduce risk and meet required safety standards
- **Comfort & Fit:** Maximum comfort, minimum weight, proper size/adjustment → encourages use
- **Freedom of Movement:** Must not restrict essential movements → prevents accidents, improves productivity
- **Durability & Maintenance:** Strong construction, easy to clean and maintain, long service life
- **Compliance with Standards:** National/international safety standards → certified protection
- **Workplace Environment:** Temperature, humidity, dust, chemicals, electrical hazards → e.g., heat-resistant clothing in hot areas
- **Compatibility:** Must not interfere with other PPE (e.g., goggles should fit under helmet)
- **User Acceptance:** Attractive & acceptable to workers → improves compliance
- **Cost Effectiveness:** Economical purchase, maintenance & replacement costs (safety is never compromised)

Q10 – Need & Significance of PPE (Same as Q8 – See Q7 & Q8)

- **Same core structure as Q8:** Need → Significance (body-part by body-part) → Advantages → Characteristics
- **Advantages:** Reduces accidents/injuries, prevents occupational disease, improves morale & productivity, lowers medical/compensation costs

Q11 – PPEs Commonly Used by Labour Force

- **Head:** Safety helmets (Class G, E, C)
- **Eye & Face:** Safety glasses, goggles, face shields, welding shields
- **Respiratory:** N95 masks, dust masks, air-purifying respirators, SCBA
- **Hands:** Nitrile, latex, Kevlar, rubber insulating gloves
- **Hearing:** Ear plugs, ear muffs
- **Feet:** Safety boots, steel-toed boots, slip-resistant shoes, metatarsal guards
- **Body:** Aprons (leather/PVC), high-vis vests, full-body protective suits
- **Fall:** Safety belts, harnesses, lifelines, lanyards

Q12 – PPE – Use and Abuse at Workplaces

Uses

- **Head:** Helmets – falling objects & impact
- **Eye & Face:** Goggles/shields – dust, flying particles, chemical splashes
- **Respiratory:** N95/respirators – dust, fumes, vapors, pathogens
- **Hands:** Gloves – cuts, burns, chemicals, electrical hazards
- **Hearing:** Ear plugs/muffs – noise, prevents hearing loss
- **Feet:** Safety boots – crushing, punctures, slips
- **Fall:** Belts & lifelines – heights above 2 m

Abuse / Misuse

- **Failure to wear:** Carelessness or discomfort → no protection
- **Improper fit:** Oversized/undersized PPE reduces protection
- **Using damaged PPE:** Cracked helmets, torn gloves, damaged respirators
- **No maintenance:** Failure to clean, inspect, and maintain regularly
- **Incorrect selection:** Wrong PPE type for the hazard
- **Modification:** Altering PPE reduces its effectiveness
- **Sharing contaminated PPE:** Spreads infections
- **Ignoring inspection:** Safety belts/lifelines not inspected → may fail

Prevention of Abuse

- Training, correct selection & fitting, regular inspection, replace damaged PPE, enforce safety rules, awareness programs, monitor compliance

Q13 – Necessary Actions at Workplace

- **Risk Assessment:** Identify hazards, assess risk level, corrective action, continuous monitoring
- **Engineering Controls:** Ventilation, humidity <60%, splash guards, isolate hazardous processes
- **Administrative Controls:** Safety policies, employee rotation, limit exposure, health monitoring
- **Hygiene & Housekeeping:** Cleanliness, disinfection, safe water, waste disposal
- **Training & Education:** Hazard recognition, first aid/CPR, hygiene practices, safety awareness
- **Vaccination & Surveillance:** Vaccines, periodic medical exams, monitor employee health
- **PPE:** Helmets, gloves, goggles, respirators, ear plugs, safety shoes
- **Emergency Preparedness:** Response plans, drills, emergency teams, first aid kits
- **Check–Call–Care:** Check scene/victim → Call emergency services → Care (first aid/CPR)
- **Mental Health:** Reduce overload, clear roles, counseling, work-life balance

Q14 – Environmental Management Plan (EMP) for Safety & Sustainability

Objectives

- Protect health & safety, prevent pollution, conserve resources, ensure regulatory compliance, promote sustainability, reduce disaster risks

4 Stages of EMP

- **Conceptualization:** Preliminary assessment, identify impacts, consider in project design
- **Planning:** Detailed impact studies, design safeguards, allocate resources
- **Execution:** Implement protection measures, install pollution control equipment, enforce OHS
- **Operation:** Monitor performance, evaluate effectiveness, take corrective action

EMP for Safety

- **Occupational Health & Safety:** Adopt OHS standards, provide PPE, safety training, hazard monitoring
- **Emergency Management:** Documented procedures, trained employees, drills, emergency coordinator, defined roles
- **Response Standards:** Life-threatening: within 3–4 min; Non-life-threatening: first aid within 15 min
- **Check–Call–Care:** Check → Call → Care
- **Disaster Planning:** Fire, explosion, gas leak, earthquake, flood – firefighting equipment, safety appliances, training

EMP for Sustainability

- **Liquid Effluent:** Treat before discharge, prevent groundwater contamination, water reuse
- **Air Pollution:** Emissions within limits, install control equipment, monitor air quality, adequate stack height
- **Solid Waste:** Suitable disposal sites, no groundwater contamination, safe reactive waste disposal, tree plantation
- **Noise & Vibration:** Reduce industrial noise, control machinery vibration
- **Housekeeping:** Cleanliness inside & outside industry, proper waste collection
- **Transport:** Parking facilities, road safety signs, prevent congestion
- **Waste Recovery:** Recycle materials, reuse treated wastewater for non-edible crops
- **Green Belt:** Tree planting around premises, maintain green belt to reduce pollution
- **Environmental Management Cell:** Dedicated cell with trained personnel for planning & implementation

13 EMP Components (Frequently Asked)

- 1. Liquid Effluents 2. Air Pollution 3. Solid Wastes 4. Noise & Vibration 5. OHS 6. Prevention/Maintenance of Control Systems 7. Housekeeping 8. Human Settlements 9. Transport Systems 10. Recovery & Reuse 11. Vegetal Cover 12. Disaster Planning 13. Environmental Management Cell

Q15 – EMP for Safety (Brief / 5 Marks)

- **Occupational Safety & Health:** OHS standards, hazard identification, PPE, safety inspections, health monitoring
- **Emergency Management:** Documented procedures, competent employees, drills, defined responsibilities
- **Life-Threatening:** Response within 3–4 min
- **Non-Life-Threatening:** First aid within 15 min
- **Check–Call–Care:** Check → Call → Care (first aid/CPR)
- **Disaster Planning:** Fire, explosion, gas leak, earthquake, floods; firefighting equipment ready; training
- **Environmental Management Cell:** Trained personnel for implementation & monitoring

Q16 – PPEs at Engineering Industries & Municipal Solid Waste (MSW) Sites

Engineering Industries

- **Head:** Safety helmets
- **Eye & Face:** Safety goggles, face shields, welding shields
- **Respiratory:** N95 masks, respirators
- **Hands:** Nitrile, rubber, cut-resistant gloves
- **Hearing:** Ear plugs, ear muffs
- **Feet:** Safety shoes, steel-toed boots
- **Body:** Leather/PVC aprons, protective clothing, high-vis vests
- **Fall:** Safety belts, harnesses, lifelines

MSW Sites

- **Head:** Safety helmets
- **Eye & Face:** Safety goggles, face shields
- **Respiratory:** Dust masks, N95 respirators
- **Hands:** Heavy-duty rubber gloves, chemical-resistant gloves
- **Feet:** Safety boots, steel-toed gum boots

- **Body:** Protective overalls, aprons, high-vis jackets
- **Hearing:** Ear plugs (where machinery is used)
- **Fall:** Safety belts & harnesses (elevated locations)
- **MSW-Specific Risks:** Cuts, tetanus, skin infections, respiratory diseases, vector-borne diseases from contaminated waste